

1. Abstract
2. Introduction
3. Literature Review
4. Materials and datasets
 - a. Dataset used - Cassini

The main experiments for this thesis were run on the dataset used in the paper [Cassini et. al.], where they focused on the semantic segmentation and object detection over the Mesopotmian Floodplain in Iraq. The dataset included 5025 images and masks. The dataset was collected using Bing maps and CORONA data imported into QGIS. The paper evaluated the results using IoU, Precision, Recall, MCC, which are defined as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN},$$

$$Recall = \frac{TP}{TP + FN}; Precision = \frac{TP}{TP + FP},$$

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}.$$

This paper compared U-Net versus MA-net, Resnet18 versus Efficientnet-B3 and Dice Loss versus Focal Loss before finally settling on the best model - MA-net, Efficientnet-B3 and Focal Loss, trained for 20 epochs

- b. Models used:
 - i. Segformer: <https://github.com/NVlabs/SegFormer>
Implementation: NVlabs
 - ii. MA-net:
<https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9201310>
Implementation: segmentation_model in pytorch(?)
 - iii. DeepLab V3+: <https://arxiv.org/pdf/1606.00915> Implementation:
segmentation_model in pytorch(?)
- c. .tar file format - why is it used and how

5. Methodology
 - a. Workflow of the PADS profiler
 - b. Metrics captured

6. Results
7. Ablation study
8. Conclusions
9. Future work
10. References